

Che-Sang Park

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Homepage

Research Interests

Robot Motion Planning with Generative Models
Physical AI, Robot Foundation Model, VLA
Reinforcement Learning, Generative Models

Education

Seoul National University

Ph.D. in Mechanical Engineering

Robotics Laboratory (<https://sites.google.com/robotics.snu.ac.kr/fcp/>)

Advisor: Prof. Frank Chongwoo Park

Seoul, Republic of Korea

Sep 2020 - Present

Seoul National University

B.S. in Mechanical Engineering

B.S. in Computer Science and Engineering

Graduated *cum laude* (GPA: 3.86/4.3)

Seoul, Republic of Korea

Mar 2013 - Aug 2020

Mar 2018 - Aug 2020

University of Washington

Exchange Student

Seattle, WA, USA

Sep 2019 - Dec 2019

Publication

[P1] Spatial Attention: Adapting Execution Horizons for Diffusion Policies via Observation Sensitivity

Che-Sang Park, Junsu Ha, Jianlong Fu, Frank C. Park

arXiv, 2026

[C2] CompVLA: A Variable Compliance Vision-Language-Action Model for Contact-rich Manipulation

Jongmin Kim*, Junsu Ha*, **Che-Sang Park***, Minchang Song, Hyeokju Jeong, Himchan Hwang, Jianlong Fu, Frank C. Park

CoRL 2026

[C1] Value Gradient Sampler: Learning Invariant Value Functions for Equivariant Diffusion Sampling

Himchan Hwang, Hyeokju Jeong, Dong Kyu Shin, **Che-Sang Park**, Sehee Kweon, Sangwoong Yoon, Frank C. Park

AISTATS, 2026

[J1] Diverse Policy Learning via Random Obstacle Deployment for Zero-Shot Adaptation

Seokjin Choi*, Yonghyeon Lee*, Seungyeon Kim, **Che-Sang Park**, Himchan Hwang, Frank C. Park

IEEE Robotics and Automation Letters (RA-L), 2025

Research Experience

Microsoft Research Asia

Intern (Full-time, Mentor: Jianlong Fu)

- Developing Robust and Efficient Diffusion Policy by applying Attention

Beijing, People's Republic of China

Sep 2024 - Mar 2025

SDU Elite Robot Summer School 2022

Participant

- Networking with other international robotics students

Odense, Denmark

Aug 2022

Music & Audio Research Group

Intern (Full-time)

- Lyric Alignment & Face Generative models using human voices

Gwanggyo, Republic of Korea

Jun 2019 - Aug 2019

Biorobotics Laboratory

Intern(Full-time)

- Soft robot grasping control with deep neural network

Seoul, Republic of Korea

Mar 2018 - Nov 2018

Projects

Hierarchical Robot Foundation Model with Motion Description Language

Project Co-Leader

- Co-leading of motivation definition, model structure implementation

Seoul, Republic of Korea

July 2025 - Present

Bringing Attention into Diffusion Policy:

Adaptive Execution Horizon for Efficient and Responsive Control

Project Leader (Advising: Jianlong Fu)

- Took the lead in defining the motivation, developing the core idea, and conducting experiments
- Achieved a 20% performance increase over Diffusion Policy under the same computation cost

Seoul, Republic of Korea

Sep 2024 - Present

Project Manager

- Integration of research efforts from six distinct groups into a cohesive, real-world robotic task scenario

ManiSkill Challenge (<https://sapien.ucsd.edu/challenges/maniskill2021/>)
Team Leader

Online
Jul 2021 - Jan 2022

- Lead the team to apply offline RL algorithms to 7-DOF robots to conduct complex manipulation tasks

Teaching & Advising

[M2794.00600 001] Optimization and Machine Learning	Spring 2026
Head TA	
[M3228.002300 001] Geometric Methods for High-Dimensional Data Analysis	Fall 2025
Head TA	
[M2794.009600 002] Advanced Topics in Mechanical Engineering 1 (Robot Vision)	Fall 2023
TA	
[M2794.001200 001] Dynamics	Fall 2021
TA	
[M2794.002700 001] Introduction to Robotics	Spring 2021
TA	

Skills Summary

Languages: Korean(Native), English(GRE: V 154 Q 169 W 3.0 (Mar.23.2019), TOEFL: 102/120 (Sep.29.2018))
Programming Languages: Python, ROS, C++, Java, MATLAB, Bash
Tools: Git, PyTorch, TensorFlow, Linux/Ubuntu, Vim, VSCode, Latex

Honors and Awards

BK21 Fellowship	Aug 2020 - Present
Full Semester Honor Scholarship	Aug 2014 / Mar 2015 / Aug 2015 / Aug 2018
Half Semester Honor Scholarship	Feb 2016